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Does Rubric-Guided Refinement Improve Automatically Optimized Prompts? A Pre-Specified Negative-Result Study

LINH NGUYEN[1], QUANG-VINH DANG[2], MINH NGOC DINH[3], AND THUY NGUYEN[1]
[1]School of Science, Engineering & Technology, RMIT University Vietnam, 702 Nguyen Van Linh Boulevard, Ho Chi Minh City 700000, Vietnam
[2]British University Vietnam, Hung Yen, Vietnam
[3]Millenia Education, Ho Chi Minh City, Vietnam

Corresponding author: Quang-Vinh Dang (e-mail: vinh.dq4@buv.edu.vn)

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ABSTRACT Meta-Prompted Instruction Refinement (MPIR) is a post-hoc layer that asks a language model to score an automatically optimized prompt against a seven-criterion rubric, rewrite it, and keep the rewrite only if it beats the original on held-out questions. An earlier evaluation with a closed refiner reported mixed gains, but audits found a validation leak and an unscored baseline, and a small pilot left its refinement rounds, statistics and answer-extraction harness open to criticism. We report a re-test whose decision rules were fixed in a time-stamped protocol. We repaired the refinement loop, logged every round, and refined 45 base prompts (PromptWizard, APE and ProTeGi on nine Big-Bench Hard tasks, up to three seeds) with Qwen3-1.7B as target and refiner. The layer changed the prompt in 1 of 80 runs and won in 0 of 45 cells, mainly because the small refiner rarely produced a usable candidate that kept the answer format. Identical prompts disagree on 6.5% of examples between scorings, and one answer-format exemplar raised the strict accuracy of exemplar-free prompts by +19 and +42 points. In an exploratory probe, a 235B-parameter refiner won in 4 of 9 cells under strict scoring, but most of these gains reverse under lenient scoring and a rubric-free rewrite reaches similar gains, so they do not isolate the rubric. Conclusions are limited to Qwen3 targets, nine tasks and the optimizers tested.

INDEX TERMS Automatic prompt optimization, Big-Bench Hard, Large language models, Meta-prompting, Negative results, Prompt engineering

1. INTRODUCTION

Large language models (LLMs) are used for translation [1], summarization [2] and question answering [3] (see [4] for a survey), and are being applied in domains such as law and medicine [5], [6]. Their accuracy depends on the prompt, and small changes in phrasing or formatting can change it substantially [7], [8]. Writing and tuning prompts is therefore a practical bottleneck for reliable deployment.

Two families of approaches address that bottleneck. Manual prompting applies heuristics distilled from practice, such as chain-of-thought reasoning [9], role assignment [10] and few-shot demonstration [11]; these are interpretable but need expertise and trial and error [12], [13]. Automatic prompt optimization (APO) searches the space of prompts with the help of an LLM, using critique, mutation, evolution or optimizer-style proposal loops [14]–[18]; it scales, but the prompts it produces are often less structured and harder to interpret than human-written ones [19]. A natural idea, which we call Meta-Prompted Instruction Refinement (MPIR), is to add a cheap post-hoc stage after any APO run: ask an LLM to score the optimized prompt against a rubric of manual heuristics, ask it to rewrite the prompt accordingly, and keep the rewrite only if it scores better on held-out validation questions.

Whether such a layer helps matters in practice. It costs a few model calls and can be added to any APO pipeline, but if it does not help those calls are wasted, and an ungated version can make prompts worse. A clean test is not trivial: the rewrite is produced by a model that may be no more capable than the target, the

gate uses a small validation set, and benchmark accuracy depends on how answers are extracted. We therefore report a pre-specified test, and we treat a null result as informative.

The idea was first evaluated with GPT-4o as the refining model and GPT-3.5-turbo as the target on the tasks of Big-Bench Hard (BBH) [20]. That evaluation reported per-task gains over PromptWizard [18] but a pooled gain that was not significant (Appendix D reproduces its numbers). A later audit found that its validation stage scored candidates on the optimizer's own training examples and that its loop never scored the unrefined prompt. An earlier version of this study was submitted elsewhere and was not externally reviewed; the first evaluation's numbers are exploratory and are not used as evidence here. A pilot with Qwen3-1.7B in every role then found no effect, but its seven refinement rounds were not independent attempts, and its statistics, task choice and answer-extraction harness were open to criticism. This paper replaces the pilot with a new study.

We ask four research questions. RQ1: with the refinement loop repaired so that rounds are genuinely distinct attempts, does gated MPIR improve a base optimizer's prompt over many tasks, optimizers and seeds? RQ2: is a failure to improve a property of the candidates or of the coarse validation gate? RQ3: does a more capable refining model change the answer? RQ4: does the rubric judge measure anything related to accuracy? (Terms such as cell and run are defined in Section 4.2.) In brief: with a 1.7B refiner the layer changed the prompt in 1 of 80 runs and won in 0 of 45 cells, mainly because that refiner rarely produces a usable candidate that keeps the answer format (RQ1, RQ2); a much larger refiner obtained strict-metric gains in several cells, but most of them reverse under lenient scoring and a rubric-free rewrite reaches gains of the same kind (RQ3); and the rubric score is only weakly related to accuracy (RQ4).

The contributions are:

- A repaired, instrumented refinement loop. Rounds differ by sampling and, in the chained variant, by the history of rejected candidates; every round is logged with its outcome (fallback, echo, duplicate, validated), its validation and test scores and the judge's rubric total.
- A pre-specified, multi-seed, multi-task evaluation. 45 base prompts (three optimizers, nine tasks, up to three seeds), decision rules fixed in a time-stamped protocol, and a random, documented choice of six tasks to complement the three pilot tasks.
- Diagnostics that make the result interpretable. Oracle scoring of all candidates, acceptance-rule sensitivity, a serving-noise floor and minimum detectable effect, gated and format-protected controls, strict and lenient answer extraction, a comparison with the uncorrected algorithm and a judge-validity analysis; defects of the pipeline and the serving configuration are catalogued (Appendix C).

2. RELATED WORK

2.1 MANUAL PROMPTING HEURISTICS

Prompting steers an LLM through in-context learning without changing its weights [11], [21], and has been described as a form of natural-language programming [13]. Non-expert authors struggle because they get no structural feedback and over-fit to the few examples they try [12]. Experienced authors rely on heuristics. Few-shot demonstrations help [11], [22], though Min et al. show that format, label space and input distribution matter more than label correctness [23]. Chain-of-thought prompting decomposes multistep problems [9], [24], and coherent, relevant reasoning steps matter more than length [25], [26]. Role prompting assigns a persona to guide reasoning [10], but its benefit is inconsistent: poorly aligned personas can hurt [27], and a large study found that adding personas to system prompts does not reliably improve performance [28]. Step-back prompting abstracts key principles before solving [29]; a behavioural benchmark of 35 LLMs found that it fosters model-based behaviour in sequential decision tasks [30], although this is weaker evidence for knowledge-intensive reasoning than the original step-back results. Models attend unevenly to information in long contexts [31], [32], and vendor guidance reports recency effects for some models while reporting no difference for GPT-4-class models [33]; whether either matters for the short prompts studied here is an empirical question rather than a premise. Self-consistency, a decoding-time heuristic, samples several reasoning paths and votes [34];

2.2 AUTOMATIC PROMPT OPTIMIZATION AND META-PROMPTING

APO methods generate, evaluate and select prompts iteratively [19]. APE induces instructions from input-output demonstrations [14]; ProTeGi treats LLM critique as a textual gradient and runs a beam search [15]; EvoPrompt applies evolutionary operators [16]; OPRO conditions an LLM on a trajectory of scored prompts [17]; PromptWizard combines critique-based refinement, example selection and self-generated reasoning [18]. DSPy, TextGrad, Self-Refine and Reflexion generalize the idea of optimizing or refining text with LLM feedback [35]–[38]. Meta-prompting supplies an LLM with a template that structures the critique (see [39] for a survey): PE2 uses a step-by-step diagnosis template [40], dual-phase methods add a meta-instruction-guided initialization [41], and PROPEL injects expert-derived principles as priors [42]. Closest in spirit to MPIR, a two-stage approach for few-shot relation extraction first searches and then locally edits the winner [43]. More recent work explores evolutionary reflection [44], error taxonomies [45], format optimization [46], debate-

based fitness [47] and RL over edits [48], and several studies target small or open-weight models: gradient-guided optimization with small models [49], minimal-edit optimization for on-device models [50], and error-driven optimization of arithmetic reasoning [51]. Others distill reasoning behaviour into prompts [52], regularize prompts for token efficiency [53], question what drives optimizer gains [54], or codify principled instructions [55].

2.3 LIMITS OF SELF-REFINEMENT AND OF LLM JUDGES

MPIR asks a model to critique and rewrite its own prompt, which places it in the self-refinement family [37], [38]. Evidence on intrinsic self-correction is sobering: LLMs cannot reliably self-correct reasoning without external feedback [56], and small models need strong verifiers to benefit from correction [57]. MPIR's critic is also an LLM judge, and LLM judges show position, verbosity and self-preference biases [58]–[60]. Finally, benchmark accuracy depends on prompt format and on how answers are extracted [8], [61], [62], which matters here because MPIR's gate and our metrics both rely on a delimiter-based extraction. These strands predict that a 1.7B model refining its own prompts may fail for reasons unrelated to the rubric, which is why our design varies the refiner, validates the judge and re-scores answers leniently.

2.4 REPORTING NEGATIVE RESULTS

Reproducibility programmes [63] and guidance on reporting experimental results [64] call for budgets, variance and full protocols. Variance across seeds and data splits can exceed the effect of a method [65], significance testing in NLP needs care [66], and small test sets leave studies under-powered [67]. We follow this guidance by fixing decision rules in advance, reporting per-cell paired differences with intervals rather than tests that cannot be informative, and quantifying the noise floor and the minimum detectable effect.

2.5 POSITIONING

Table 1 contrasts how the methods above propose and select candidates. It states design differences, not evidence of superiority: whether any of them outperforms the others is not tested here.

What distinguishes MPIR in this table is its role, not a mechanism: it adds one more critique-rewrite-validate cycle on top of whatever an upstream method produced. That makes it cheap and optimizer-agnostic, but also means its mechanism is close to existing critique-and-revise loops [15], [37], [40]; we do not claim novelty beyond the combination of an explicit rubric with a post-hoc, gated role.

TABLE 1: Design comparison of MPIR with representative methods. The table describes design differences only; it reports no empirical comparison.

Method	Candidate generation	Selection	Role relative to an APO run
APE [14]	Induction from demonstrations	Execution accuracy	Optimizer
ProTeGi [15]	Textual-gradient edits	Beam search on minibatches	Optimizer
EvoPrompt [16]	Crossover and mutation	Fitness per generation	Optimizer
OPRO [17]	LLM proposal given score history	Best of trajectory	Optimizer
PromptWizard [18]	Critique and sequential re-optimization	Scores on training batches	Optimizer
PE2 [40]	Structured diagnosis template	Held-out scoring	Optimizer
PROPEL [42]	Expert principles as priors	Implicit in the priors	Optimizer
GEPA [44]	Evolution with reflection	Reflection-guided selection	Optimizer
MPIR (tested here)	Rubric-guided critique and rewrite	Validation accuracy vs. the unrefined prompt	Post-hoc layer

3. THE MPIR LAYER UNDER TEST

MPIR is a three-stage cycle applied to a prompt P0 that an upstream APO method has already produced: (1) a language model scores P0 against a seven-criteria rubric of manual prompting heuristics and recommends improvements; (2) it rewrites P0 accordingly; (3) the rewrite is scored on validation questions and kept only if it beats the incumbent (Figure 1). We describe it here as the object under test, not as a contribution.

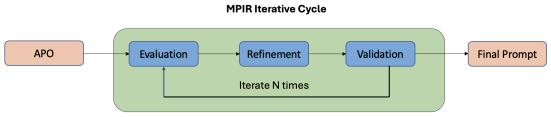


FIGURE 1: The MPIR cycle as drawn for the first submission: a prompt P0 from an APO run enters evaluation, refinement and validation for N rounds, and the best validated prompt is returned. In the legacy loop every round starts from P0; the repaired variants of Section 3.2 sample the refiner and, in the chained variant, start each round from the incumbent prompt.

3.1 PROBLEM FORMULATION

Let (Qv, Av) = {(qi, ai)}, i = 1..M, be a validation set of M question-answer pairs and A(L, P) the accuracy of the target model L on it under prompt P. Let R(P0) be the set of candidate prompts that the refinement loop produces from P0. MPIR returns

Pfinal = argmax over P' in {P0} ∪ R(P0) of A(L, P'), with ties resolved in favour of P0 (a candidate must be strictly better). Including P0 in the candidate set is what makes it possible for the procedure to conclude that no refinement helps.

3.2 ALGORITHM AND VARIANTS

Algorithm 1 is the repaired loop. It scores P0 first, so the incumbent score Abest starts at A(L, P0) instead of minus infinity. In the original implementation the incumbent score started at minus infinity, so the first round's candidate was accepted whatever it scored, which is the defect Section 5.9 quantifies. Choosing among {P0} ∪ candidates by held-out score is standard in APO loops, so we regard this as an implementation defect of ours, not as a general property of the method.

Algorithm 1 MPIR with a validation gate (repaired)

Require: L: target LLM; L': refining LLM; Meval, Mrefine: meta-prompts; R: rubric; P0: prompt from an APO run; (Qv, Av): validation set; N: rounds; mode in {independent, chained}; τ: sampling temperature
Ensure: Pfinal

```
Pbest ← P0; Abest ← Validate(L, P0, (Qv, Av)); H ← empty list   ▷ score the baseline first
for t = 1 to N do
    Psrc ← P0 if mode = independent else Pbest
    S ← Evaluate(Meval, L', Psrc, R; τ)   ▷ rubric critique
    P' ← Refine(Mrefine, L', Psrc, S, H; τ)   ▷ H is empty when independent
    outcome ← Classify(P')   ▷ fallback, echo, duplicate or validated
    if outcome = validated then
        A' ← Validate(L, P', (Qv, Av)); append (P', A') to H
        if A' > Abest then Pbest ← P'; Abest ← A'   ▷ strict: a tie keeps the incumbent
    end if
end for
return Pbest
```

The pilot ran what we call the legacy loop: Psrc = P0 in every round, H empty and τ = 0. Because Evaluate and Refine then receive identical input in every round, the seven rounds were one request repeated, up to serving non-determinism, and 'seven rounds without acceptance' meant roughly one distinct attempt per cell. The repaired loop differs between rounds in two ways: independent mode samples at τ = 0.7 so each round is a fresh draw, and chained mode refines the current incumbent and shows the model the earlier candidates with their validation scores, as OPRO does with its trajectory [17]. We study seven variants (Table 2). Two of them, V1P and V2P, are format-protected: the answer-format line is appended to every candidate, as it is for the rubric-free rewrite and for the prompts of the base optimizers. They were added after an audit found that the refinement request does not ask the refiner to keep the answer-format line, which the 1.7B refiner then dropped in most candidates (Section 5.2); they are exploratory.

TABLE 2: Refinement variants studied. All use the same gate (strict improvement over the incumbent on 25 validation questions) and N = 7 rounds. V3 candidates, and those of V1P and V2P, carry the answer-format line; V1 and V2 candidates are scored as the refiner wrote them.

Variant	Refines	History	Sampling	Rubric text
V0 legacy (as in the pilot)	P0	none	greedy	original
V1 independent	P0	none	τ = 0.7	corrected
V2 chained (primary)	incumbent	scored candidates	τ = 0.7	corrected
V2L chained	incumbent	scored candidates	τ = 0.7	original
V3 gated free rewrite	P0	none	τ = 0.7	none (no rubric)
V1P independent, format-protected	P0	none	τ = 0.7	corrected
V2P chained, format-protected	incumbent	scored candidates	τ = 0.7	corrected

3.3 THE SEVEN-CRITERIA RUBRIC

The rubric was assembled from the prompting literature [68], [69], industry guidance [33], [70]–[72] and iterative experimentation on BBH-style tasks. It scores a prompt from 1 to 5 on: (1) Role Prompting, (2) Step-Back Reasoning, (3) Guided Chain of Thought, (4) Instruction and Separation, (5) Output Format with Examples, (6) Worked Reasoning in Examples and (7) Conclusion, for a total out of 35. The evidence behind the criteria is uneven. Chain-of-thought and demonstrations are well supported [9], [23], [26]. Role prompting is the least supported, since its benefit is inconsistent [27], [28]. The Conclusion criterion extrapolates positional-bias results from long contexts [31], [32] to short prompts, and the vendor page we cite for it states that instruction placement made no difference for GPT-4-class models [33]. Industry sources are practitioner guidance rather than empirical evidence; they are living documents, so

the reference list gives archived copies and access dates. The rubric was refined during the first evaluation using BBH tasks, including the three Tier-1 tasks, so Tier 1 is not independent of the rubric's development; the six random Tier-2 tasks are.

The rubric text used in the pilot was internally inconsistent: it announced 'five defined criteria', asked for 'all 6 criteria' and 'Total Score: X/25' while listing seven criteria. We found this during the audit. The primary variants use a corrected text (seven criteria, X/35); variant V2L keeps the original text as a sensitivity check.

3.4 EVALUATION, REFINEMENT AND OUTPUT GUARDS

The evaluation meta-prompt (Figure 2) asks the refining model to apply the rubric to the prompt and report scores, strengths, weaknesses and 7-10 suggestions. The refinement request (Figure 3) asks for a rewrite wrapped in <START> and <END>; in chained mode it also lists the earlier candidates with their validation accuracy and asks the model not to repeat them. Three output guards exist, and each outcome is logged per round. A response without delimiters is a fallback: it cannot be validated, and the pilot's code returned P0 in that case, which ties the baseline by construction. A candidate containing phrases verbatim from the evaluation template, or APE's induction template, is an echo and is rejected before validation. A candidate textually identical to the incumbent, to P0 or to an earlier candidate is a duplicate. Only the remaining candidates are validated. The refinement request of V1 and V2 does not ask the refiner to keep the answer-format line (the free-rewrite prompt does), and a candidate is scored exactly as written; this asymmetry matters for small refiners and is examined in Section 5.2.

3.5 THE VALIDATION GATE

A candidate is accepted only if its accuracy on the 25 validation questions strictly exceeds the incumbent's. With 25 questions accuracy moves in steps of 4 percentage points, and one question of serving noise equals a step, so acceptance is rare by design. Whether the gate or the candidates explain the absence of accepted refinements is research question RQ2, which we answer by scoring every candidate on the full test split (the oracle analysis) and by re-applying other acceptance rules offline (Section 5.4).

4. EXPERIMENTAL DESIGN

4.1 PRE-SPECIFICATION AND DECISION RULES

The pilot's go/no-go rule was written in the project's planning document (REBUILD.md, commit 7612aa0, 2026-09-03) before the pilot ran. That is a time-stamped plan in a version-controlled repository, not an entry in a public registry, so we describe the study as pre-specified, not pre-registered; several pipeline fixes were also made while the pilot ran (Appendix C). The chronology is:

```
### Task Overview
You are a Senior Prompt Engineer with
two main tasks:
1. Evaluate prompts using a
7-criteria rubric. 2. Refine prompts
by applying evaluation feedback.
### Prompt Under Evaluation
[Prompt]: {prompt}
### Instructions
1. Review Prompt 2. Apply Rubric
(seven defined criteria)
3. Score & Justify: score 1-5, one
strength, one weakness, a rationale
per criterion
4. Calculate Total: cumulative score
out of 35
5. Recommend 7-10 actionable
improvements 6. Report in the
standardized format
### Seven-Criteria Evaluation Rubric
1. Role Prompting 2. Step Back
3. Guided Chain of Thought 4.
Instruction & Separation
5. Output Format with Examples 6.
Worked Reasoning in Examples 7.
Conclusion
### Output Format
1. Role Prompting - X/5 - Strength:
... - Improvement: ... - Rationale:
...
(continue for all 7 criteria) **Total
Score: X/35** Refinement Summary: ...
```

FIGURE 2: Evaluation meta-prompt (corrected text; abridged from the repository file).

```
Refine the prompt by applying all
suggestions from the evaluation.
Make sure to wrap the refined prompt
with <START> and <END>

[chained mode adds: Earlier
refinement attempts ... did NOT beat
the current best prompt on the
validation questions; do not repeat
them: Attempt k (validation accuracy
a vs current best b): ...]
```

FIGURE 3: Refinement request used by the independent (top) and chained (bottom) variants.

first evaluation and audit (before September 2026); pilot, 2026-09-04 and 05; first submission to this journal, 2026-09-05; decision and reviews, 2026-10-19; revision protocol committed 2026-10-19 (commit 8fd1397), before any revised experiment; revised experiments, 2026-10-20 and 21. The decision rules below are reproduced from that protocol. Its dated amendments, which include one change of the refiners used for D2 and later exploratory additions, are listed with their dates in Appendix F.

- D1 (RQ1). Primary outcome per cell: paired difference in test accuracy between the final MPIR prompt and P0 (10,000-resample paired bootstrap,

95% percentile interval); a cell is a win if the difference is positive and the interval excludes zero. The effect 'reproduces' if at least 15 of 45 cells are wins and the mean cell difference is at least +2.0 points; it is 'null' if at most 2 of 45 are wins and the mean lies within ±1.0 point; otherwise 'mixed'. The protocol did not say how the up to three MPIR seeds of a cell are combined: we pool them by averaging the per-example outcomes of the final prompts over the seeds before the bootstrap. The planned 135 runs (45 cells times three MPIR seeds) were cut to 80 for compute reasons, so cells have one to three runs.

- D2 (RQ3). Capability dependence is supported if the mean cell difference with the strongest refiner exceeds that with the 1.7B refiner by at least 2.0 points and at least 3 of 9 cells are wins with that refiner. The protocol named a local 8B refiner; because the shared GPU was too slow for the extra servers, the strongest-refiner role was reassigned on 2026-10-20 (before any result with these refiners was inspected) to GPT-6 Luna, with Qwen3-235B-A22B as an intermediate rung. Any statement about Qwen3-235B against D2 is therefore post hoc.
- D3 (RQ2). The gate is limiting if the best oracle candidate beats P0 on test by at least 3.0 points in at least 3 of 9 cells although the gated final prompt did not.

McNemar, GEE and Wilcoxon tests on identical prompts cannot be informative, and the Wilcoxon test on three tasks cannot go below $p = 0.25$, so none is reported as inference. Rubric-accuracy correlations, lenient parsing, the uncorrected-algorithm comparison, the noise floor and the minimum detectable effect are descriptive.

4.2 DATA, SPLITS AND TASK SELECTION

BBH has 23 tasks in 27 task files, about 6,511 examples [20]. Each task is split three ways with random.Random(seed).sample over the original example order: 25 optimizer-training examples, 25 validation examples for the gate, and the remainder as test (200 examples for every task used here except penguins_in_a_table, which has 96). The partition is stored with the original example indices, is shared by all optimizers and conditions for a given (task, seed), and no stage touches test data during optimization or refinement. Tier 1 is the three pilot tasks (hyperbaton, ruin_names, penguins_in_a_table) at seeds 42, 43 and 44. They were chosen because the leaked first evaluation showed the largest gains there, so they are a biased sample and may be subject to winner's-curse effects. Tier 2 is six tasks drawn uniformly at random, without replacement, from the other 24 task files with random.Random(20261019): temporal_sequences, movie_recommendation, disambiguation_qa, geometric_shapes, salient_translation_error_detection and sports_understanding, at seed 42. In total there are 45 cells, each with its own base prompt P0.

- Cell. A (task, optimizer, split seed) combination with its own split and its own base prompt P0.
- Run. One execution of a refinement variant on a cell with one MPIR seed (the seed of the refiner's sampling); a cell has one to three runs.
- Variant, round. A refinement procedure (V0 to V3, V1P, V2P, V2L; Table 2); one round is one evaluate-refine-validate step, with $N = 7$ rounds per run.
- Split seed, MPIR seed. The seed that draws the train, validation and test split and the optimizer's own randomness (42, 43, 44), and the seed of the refiner's sampling (1 to 3).

4.3 MODELS AND SERVING

The target model is Qwen3-1.7B [73] with thinking disabled and greedy decoding, at most 1024 new tokens. The refining model is the same model in the main grid. In the capability experiment the refiner is Qwen3-235B-A22B (the 2507 instruct release, a mixture-of-experts model from the target's family) or GPT-6 Luna [74], both reached through the OpenRouter API on 2026-10-20 and 21 with temperature 0.7, a 3000-token limit and, for GPT-6 Luna, minimal reasoning effort. Meta-prompt calls allow 1500 (evaluation) and 3000 (refinement) new tokens. The target and the 1.7B refiner are served with vLLM [75] 0.28.0 under WSL2 on one 16 GB RTX 5000 Ada GPU, bf16 weights, a maximum context of 8192 tokens (Appendix C). Greedy decoding is not bit-reproducible under continuous batching, so we measure the resulting noise (Section 5.7).

4.4 BASE OPTIMIZERS AND CONDITIONS

APE and ProTeGi are clean-room implementations of the published algorithms [14], [15], run with few_shot_count = 0, hence exemplar-free; ProTeGi scores every successor on a full minibatch instead of using the original bandit scheme. PromptWizard [18] is run with synthetic-example generation disabled; its three in-context examples are drawn from the 25 training examples with model-generated reasoning, and its generated expert identity is not used at evaluation. Hyperparameters are in Table 3. Many of the APE and ProTeGi base prompts are not optimized in any strong sense: for two of the three pilot tasks they are the seed instruction "Let's think step by step." plus the answer-format line, and the APE prompt for hyperbaton is its own induction template (Appendix A). PromptWizard is a reduced configuration, because synthetic-example generation was disabled. The base prompts are therefore a mix of weak and degenerate starting points, which limits what refining them can show. Only the expert few-shot condition uses the three BBH-author exemplars [20]; the first manuscript's statement that APE and ProTeGi also

used them was wrong. Reference conditions are zero-shot CoT [24] and expert few-shot CoT. The free-rewrite control (Appendix A.3) rewrites P0 with no rubric; V3 adds the gate.

TABLE 3: Hyperparameters of the base optimizers and of MPIR as actually used.

Component	Parameter	Description	Value used
PromptWizard	mutate_refine_iterations	Outer rounds of mutate-then-refine over the task description	3
PromptWizard	mutation_rounds	Style-mutation rounds inside one outer round	3
PromptWizard	style_variation	Prompt variants generated per call	5
PromptWizard	refine_task_eg_iterations	Critique rounds on instruction and examples	3
PromptWizard	few_shot_count / seen_set_size	In-context examples / training examples	3 / 25
APE	num_candidates, iterations	Induced candidates; resampling rounds	10, 2
APE	num_scoring_examples, top_n	Scoring minibatch; candidates kept	5, 3
ProTeGi	beam_width, num_steps	Beam width; expansion rounds	3, 3
ProTeGi	minibatch_size, eval_batch_size	Error minibatch; scoring batch	5, 8
MPIR	N, validation size	Rounds; gate questions	7, 25
MPIR	refiner temperature, max tokens	Sampling; evaluation / refinement	0.7, 1500 / 3000
MPIR	history kept (chained)	Last candidates shown; characters each	4, 800
MPIR	guards	Fallback, echo (5 phrases), duplicate	see Section 3.4

4.5 METRICS

Accuracy is the share of test examples whose delimited answer equals the gold answer after removing parentheses, whitespace and case (strict extraction, as in the pilot). Because strict extraction penalizes correct reasoning that is wrapped badly, we add a lenient accuracy computed offline from the saved raw outputs: the response counts as correct if the delimited span (or, if there is none, the last 300 characters) names exactly one option and it is the gold one. Both are reported; strict extraction remains primary because MPIR's gate uses it.

4.6 STATISTICAL ANALYSIS

For every cell we report the paired difference in test accuracy between the final MPIR prompt and P0 with a 95% paired-bootstrap interval over test examples. Cells with identical prompts have a difference of zero by construction, apart from serving noise. Across cells we report counts of wins, the mean difference and a hierarchical bootstrap interval that resamples tasks and then the cells within each selected task. The noise floor is the distribution of accuracy differences between repeated scorings of the same prompt, and the minimum detectable effect at 80% power and $\alpha = 0.05$ follows from the paired discordance rate [66], [67]. Intervals use the percentile bootstrap [76] and, for single accuracies, the Wilson interval [77].

4.7 HARNESS DEFECTS AND DEVIATIONS

Running against a live model exposed defects that mocked tests could not (truncation of long responses, answer normalization, delimiter mismatches in expert exemplars, template mismatches between an optimizer's own and MPIR's evaluation harness, two echo patterns and the unscored baseline). All are described with their magnitude in Appendix C and were fixed before the numbers reported here were produced.

5. RESULTS

5.1 BASE PROMPTS AND THE SCORING HARNESS

Table 4 gives the strict and lenient test accuracy of the nine pilot base prompts (seed 42) with Wilson intervals [77] and the chance level of each task, and Table 5 the reference conditions. Several baselines are at or below chance under strict extraction, the largest being ProTeGi: 10.5% on hyperbaton, 5.0% on ruin_names and 1.0% on penguins_in_a_table. Under lenient extraction the same responses score 58.0%, 37.5% and 45.8%. The raw outputs show the cause: the model reasons to an option but wraps a whole explanatory sentence in the answer tags, which the strict rule rejects. This happens for exemplar-free prompts (APE, ProTeGi, zero-shot) and not for prompts that contain a worked example of the bare tag (PromptWizard, expert few-shot). Lenient extraction accepts everything strict extraction accepts, plus a response whose tagged span (or, without tags, last 300 characters) names exactly the gold option; it is approximate, and a prompt that scores well on it is not guaranteed to be reasoning correctly.

Two checks bear on how much of this is a harness effect. First, appending one worked format exemplar (the first BBH-author exemplar, with its bare tag) to an exemplar-free base prompt raised strict accuracy by +19.5 points on average for APE and +42.3 points for ProTeGi over 15 cells each (the paired-bootstrap interval excludes zero in 9 and 15 cells; the range is -45.0 to +72.9 points, the negative end being APE's degenerate hyperbaton prompt). Under lenient extraction the same

TABLE 4: Test accuracy (%) of the nine pilot base prompts (seed 42, Qwen3-1.7B, greedy, one scoring pass). Strict is the primary metric with a 95% Wilson interval; lenient is the secondary metric. Chance is 1 over the number of distinct gold labels. Delims is the share of responses containing both tags.

Task	Optimizer	Strict [95% CI]	Lenient	Chance	Delims %
hyperbaton	PromptWizard	59.0 [52.1, 65.6]	59.0	50.0	100
hyperbaton	APE	44.5 [37.8, 51.4]	59.5	50.0	100
hyperbaton	ProTeGi	10.5 [7.0, 15.5]	58.0	50.0	76
ruin_names	PromptWizard	63.5 [56.6, 69.9]	63.5	16.7	100
ruin_names	APE	41.0 [34.4, 47.9]	43.0	16.7	99
ruin_names	ProTeGi	5.0 [2.7, 9.0]	37.5	16.7	98
penguins_in_a_table	PromptWizard	71.9 [62.2, 79.9]	71.9	20.0	100
penguins_in_a_table	APE	15.6 [9.7, 24.2]	59.4	20.0	94
penguins_in_a_table	ProTeGi	1.0 [0.2, 5.7]	45.8	20.0	100

TABLE 5: Reference conditions, strict (lenient) test accuracy %, seed 42. Free rewrite is the ungated rubric-free rewrite of the PromptWizard prompt from the pilot.

Task	Zero-shot CoT	Expert few-shot CoT	Free rewrite (ungated)
hyperbaton	51.0 (63.0)	86.0 (89.5)	61.0 (61.0)
ruin_names	37.0 (43.0)	50.0 (50.0)	44.5 (44.5)
penguins_in_a_table	6.2 (37.5)	72.9 (72.9)	55.2 (55.2)

intervention gains +2.3 and +10.2 points, so most of the strict gain is the tag, and the exemplar's effect on reasoning is modest. Second, the same prompt does not score the same twice: the strict accuracy of the identical nine prompts, scored in the pilot and again for this study, differs by 2.6 points on average and up to 7.0 points (Section 5.7). Accuracy differences of this size are not reliably attributable to a prompt on this serving stack.

5.2 WHAT THE ROUNDS WERE

The composition of the refinement rounds is in Table 6 (all rounds of all runs; per-cell, per-round records for the primary variant are in Appendix E). The legacy loop of the pilot (V0, nine cells) produced on average 4.3 distinct candidate prompts per run although decoding was greedy: the seven rounds were not identical copies, but they differed only through serving non-determinism, not by design, so 'seven rounds without acceptance' overstated the evidence. (The pilot's own prompts were byte-identical in all nine cells; in our re-run of the same loop 1 of nine runs changed its prompt, which shows how fragile even a greedy comparison is.) In the repaired variants the rounds differ by sampling and, for V2, by history.

For the 1.7B refiner, 60% of the chained rounds (337 of 560) never reached validation because the response had no delimiters or echoed a template; of the 193 validated candidates 126 scored below the baseline, 66 tied and 1 scored higher. Most validated candidates are far worse, not tied. An audit found a large part of the reason: the refinement request does not ask the refiner to keep the answer-format line, and 93% of the validated V1 and 95% of the validated V2 candidates lack it, so they score exactly zero under strict extraction; the rubric-free rewrite, whose prompt asks to preserve the answer format, loses it in none. The unprotected numbers therefore measure a failure of the 1.7B refiner to keep the format as much as a failure of the rubric. To separate the two we added the format-protected variants V1P and V2P, which append the answer-format line to every candidate. With protection the 1.7B refiner still reaches validation in only 30% (V1P, 70 rounds) and 27% (V2P, 56 rounds) of rounds, and 5 of its 36 validated candidates beat the baseline on validation (Table 6). The protected variants were run on the nine Tier-1 seed-42 cells and finished on 5 (V1P) and 8 (V2P) of them; the rest were not completed for compute reasons. Refinement with the original, inconsistent rubric text (V2L) was no better: 0 accepted candidates in 63 rounds and 63% of rounds lost to fallbacks or echoes, so the inconsistency in the rubric text is not what limits the 1.7B refiner.

5.3 THE REPAIRED LAYER ON 45 CELLS (RQ1, D1)

With the 1.7B model refining its own prompts through the chained variant, the final prompt equals P0 in 79 of 80 runs (45 cells, one to three MPIR seeds each). Only 1 run accepted a candidate. No cell is a win: the mean paired difference over cells is -0.04 points (hierarchical bootstrap interval [-0.16, +0.00], which is almost degenerate because 44 of 45 cells are exactly zero), and 0 of 45 cells satisfy the win criterion. By the pre-specified rule (D1) the outcome is 'null'; a rule-of-three bound puts the acceptance rate of this procedure below about 4% of runs. The result does not depend on the base optimizer: PromptWizard 1 of 28 runs changed; APE 0 of 31 runs changed; ProTeGi 0 of 21 runs changed. D1 was specified and run for the unprotected V2, and Section 5.2 shows that this procedure is limited by the 1.7B refiner's inability to produce usable, format-preserving candidates; the null is therefore a statement

TABLE 6: Outcome of every refinement round (all runs of a variant pooled; refiner Qwen3-1.7B unless stated). Fallback: no delimiters. Echo: template phrase. Duplicate: same text as an earlier prompt. Validated rounds are split by validation score against the baseline (better/equal/worse). The last column counts runs whose final prompt differs from P0.

Variant	Runs	Rounds	Fallback	Echo	Dup.	Validated	Val. better/equal/worse	Accepted	Runs changed
V0 legacy	9	63	3	16	5	39	1/7/31	1	1/9
V1 independent	27	189	23	46	0	120	2/21/97	2	2/27
V2 chained	80	560	177	160	30	193	1/66/126	1	1/80
V2L original rubric	9	63	26	14	4	19	0/4/15	0	0/9
V3 free rewrite (gated)	27	189	0	0	87	102	19/23/60	18	16/27
V1P protected	10	70	8	41	0	21	5/0/16	4	3/10
V2P protected	8	56	28	13	0	15	0/0/15	0	0/8
V2, refiner Qwen3-235B	18	126	9	5	0	112	52/4/56	21	10/18
V3, refiner Qwen3-235B	9	63	0	0	19	44	15/4/25	8	5/9
V2, refiner GPT-6 Luna	9	63	6	0	0	57	3/8/46	2	1/9

about this procedure with this refiner, and not about rubric-guided refinement in general. Table 7 gives the per-task view. The tests of the pilot (McNemar, GEE, Wilcoxon on three tasks) are not reported, because with identical prompts they cannot be informative.

TABLE 7: Chained variant, 1.7B refiner (PW = PromptWizard): in each entry, the number of runs whose final prompt differs from P0 out of all MPIR runs of that task and optimizer, then the mean paired test-accuracy difference over those cells in points. * marks the Tier-1 tasks (split seeds 42 to 44, three cells per optimizer); Tier-2 tasks have split seed 42 only.

Task	PW	APE	ProTeGi
hyperbaton *	1/7, -0.6	0/7, +0.0	0/5, +0.0
ruin_names *	0/7, +0.0	0/7, +0.0	0/5, +0.0
penguins_in_a_table *	0/5, +0.0	0/7, +0.0	0/5, +0.0
temporal_sequences	0/2, +0.0	0/2, +0.0	0/1, +0.0
movie_recommendation	0/1, +0.0	0/2, +0.0	0/1, +0.0
disambiguation_qa	0/2, +0.0	0/2, +0.0	0/1, +0.0
geometric_shapes	0/2, +0.0	0/2, +0.0	0/1, +0.0
salient_translation_error_detection	0/1, +0.0	0/1, +0.0	0/1, +0.0
sports_understanding	0/1, +0.0	0/1, +0.0	0/1, +0.0

5.4 IS THE GATE THE BOTTLENECK? (RQ2, D3)

To separate the candidates from the gate, every validated candidate of MPIR seed 1 was also scored on the full test split. Ungated, the mean test-accuracy change of a 1.7B-refiner candidate relative to its P0 is -27.0 points for the independent variant (41 candidates) and -15.2 points for the chained variant (22), and no cell has a candidate that beats P0 by 3 points or more (0 and 0 of 9 cells); these candidates mostly lack the answer-format line (Section 5.2), so the figure largely reflects format loss. Candidates that keep the format are still not better on average: -24.9 points for the format-protected 1.7B candidates of V1P and V2P (20 candidates, none beating P0 by 3 points or more), -8.8 points for the 1.7B rewrite (33 candidates) and -4.0 points for the Qwen3-235B candidates (58; best candidate per cell +4.0 points on average, 4 cells at +3 or more). D3 requires the best candidate to beat P0 by 3 points in at least 3 of 9 cells while the gated final prompt does not; it is not met for the 1.7B refiner. Validation accuracy tracks test accuracy across candidates (Spearman 0.92 over 275 oracle-scored candidates of all refiners), but this is driven by candidates that score zero; among 1.7B-refiner candidates with positive validation accuracy it is 0.37 (n = 27). The gate can therefore rank broken candidates but is a weaker guide among plausible ones, and a 25-question gate is noisy (Section 5.7). Table 8 re-applies other acceptance rules offline to the same candidates; cells with no validated candidate are omitted.

5.5 FREE REWRITING VERSUS THE RUBRIC

In the pilot, the ungated rubric-free rewrite of the PromptWizard prompt changed strict accuracy by +2.0, -19.0 and -16.7 points on hyperbaton, ruin_names and penguins_in_a_table (Tables 4 and 5); the first is 4 test items, within the serving noise of Section 5.7, and the others are large losses, so the ungated rewrite was harmful on two of three tasks. The gated rewrite (V3) accepted a rewrite in 16 of 27 runs, against 2 of 27 for the independent rubric variant and 0 of 27 for the chained one (Tier-1 runs). Part of this gap is the format asymmetry of Section 5.2, not the rewriting style. V3 has 2 winning cells (hyperbaton / APE +17.0, ruin_names / ProTeGi +2.8) and 1 cells in which the gate accepted a harmful rewrite (hyperbaton / PromptWizard -9.3), for a mean difference of +0.7 points. So the gate does not reliably protect against harm either, since 25 questions are noisy. We cannot say that unstructured rewriting is worse than rubric-guided refinement: the rubric-free variant did at least as well (+0.7 against -0.5 and +0.0 points), and the rubric adds nothing detectable with this refiner.

TABLE 8: Acceptance-rule sensitivity on the oracle-scored candidates (MPIR seed 1, Tier-1 seed-42 cells). Cells accepting: cells in which the rule accepts some candidate, out of the cells with at least one validated candidate. Mean delta is the mean change in strict test accuracy of the selected prompt versus P0, in points, over those cells (for the larger gate, measured on the held-out half of the test split). Better/worse is the number of cells with a positive or negative delta.

Variant	Acceptance rule	Cells accepting	Mean delta	Better/worse cells
V1 independent	strict (as run)	1/9	-0.9	0/1
V1 independent	non-strict (ties accepted)	2/9	-1.1	0/2
V1 independent	mean of 3 validation scorings	1/9	-0.9	0/1
V1 independent	larger gate (val + half of test)	0/9	+0.0	0/0
V2 chained	strict (as run)	0/8	+0.0	0/0
V2 chained	non-strict (ties accepted)	2/8	-0.8	0/2
V2 chained	mean of 3 validation scorings	0/8	+0.0	0/0
V2 chained	larger gate (val + half of test)	0/8	+0.0	0/0
V3 free rewrite	strict (as run)	5/9	+0.4	2/3
V3 free rewrite	non-strict (ties accepted)	7/9	-0.6	2/5
V3 free rewrite	mean of 3 validation scorings	6/9	-0.4	2/4
V3 free rewrite	larger gate (val + half of test)	3/9	+1.9	2/1
V2, Qwen3-235B	strict (as run)	5/9	+8.7	4/1
V2, Qwen3-235B	non-strict (ties accepted)	6/9	+8.6	4/2
V2, Qwen3-235B	mean of 3 validation scorings	5/9	+8.7	4/1
V2, Qwen3-235B	larger gate (val + half of test)	6/9	+7.8	4/2

5.6 A MORE CAPABLE REFINER (RQ3, D2)

With Qwen3-235B-A22B as refiner, 112 of 126 rounds (both MPIR seeds pooled in Table 6) reach validation, almost no round is an echo, no candidate lacks the answer tag, 5 of 9 first-seed runs accept a candidate, and the final prompt is a significant strict win in 4 of 9 cells (hyperbaton / PromptWizard +30.0, hyperbaton / ProTeGi +29.0, ruin_names / ProTeGi +12.0, penguins_in_a_table / ProTeGi +11.5); the mean difference is +8.7 points. A second MPIR seed gives the same pattern: 4 winning cells, mean +12.3 points. GPT-6 Luna, run with minimal reasoning, produces valid candidates in 90% of rounds but only 3 of them beat the baseline, with 1 winning cell (hyperbaton / PromptWizard +27.0) and a mean difference of +3.0 points. Under the pre-specified rule D2, with GPT-6 Luna as the strongest refiner, capability dependence is not supported: the mean exceeds that of the 1.7B refiner by +3.0 points but only 1 cell wins, against the required 3. With Qwen3-235B the strict rule is met, but only post hoc, since the role was assigned to GPT-6 Luna.

Strict accuracy is, however, the wrong judge of these gains. Under lenient extraction (Table 9) Qwen3-235B wins in 1 cell and loses in 3, with a mean difference of -3.2 points. The three ProTeGi cells in which the strict gain is large are cells where the base prompt reasons about as well as before but wraps whole sentences in the answer tags; the accepted candidate (Figure B.3) adds an explicit instruction to wrap the final answer on its own line and a worked output example, which repairs the tag, while the model's reasoning under the long structured prompt is worse (lenient accuracy falls, for example from 58.0% to 49.0% on hyperbaton with ProTeGi). A gate that uses strict extraction accepts such candidates, which is a concrete way in which answer extraction and the gate interact. The single gain that holds under both metrics is hyperbaton with PromptWizard, whose base prompt already had the tag (strict = lenient = 59.0%) and whose accepted candidates, from two different refiners, add a role, a step-back and numbered reasoning steps and reach 89% and 86%. One cell out of nine, on a base prompt that opens with an irrelevant sentence (Appendix A), is a gain that a rubric-based rewrite can plausibly claim; there is no multiplicity correction and no replication on another task. The two strong refiners differ, one belongs to the target's family and one was run with minimal reasoning, so capability, family and reasoning budget are confounded, and the experiment is an upper-bound probe, not a controlled capability scale.

5.6.1 Follow-Up Runs (Exploratory)

Because the strong-refiner signal appeared after the pre-specified analysis, two follow-up runs were added and are exploratory (Appendix F). First, the rubric-free gated rewrite (V3) with the same Qwen3-235B refiner gives a mean strict difference of +6.0 points with 2 winning cells of 9 (ruin_names / ProTeGi +26.0, hyperbaton / PromptWizard +25.5), against +8.7 points and 4 wins for the rubric-guided variant, and a lenient mean of -0.4 points. A refiner that does not use the rubric therefore reaches gains of the same kind, so the rubric is not what produces them. Second, a second MPIR seed of the rubric-guided variant gives +12.3 points and 4 winning cells, so the strict pattern is stable across seeds.

TABLE 9: Refiner capability on the nine Tier-1 seed-42 cells; the target stays Qwen3-1.7B. Validated is the share of rounds reaching validation. Runs changed, strict wins and the strict mean refer to the final prompt versus P0 under strict extraction; the last two columns give wins/losses and the mean difference under lenient extraction. 1.7B rows pool up to three MPIR seeds, the others use MPIR seed 1.

Refiner, variant	Validated	Val. better	Runs changed	Strict wins	Strict mean	Lenient wins/losses	Lenient mean
Qwen3-1.7B (self), V1	63%	2	2/27	0/9	-0.5	0/1	-0.6
Qwen3-1.7B (self), V2	41%	0	0/27	0/9	+0.0	0/0	+0.0
Qwen3-1.7B (self), V3	54%	19	16/27	2/9	+0.7	0/2	-1.8
Qwen3-235B-A22B, V2	92%	24	5/9	4/9	+8.7	1/3	-3.2
Qwen3-235B-A22B, V3	70%	15	5/9	2/9	+6.0	1/1	-0.4
GPT-6 Luna, V2	90%	3	1/9	1/9	+3.0	1/0	+3.0

5.7 SERVING NOISE AND MINIMUM DETECTABLE EFFECT

Each of the nine pilot base prompts was scored five times on the test and validation splits (identical prompt, greedy decoding). Test accuracy varies with a standard deviation of 1.4 points and a range of 3.3 points per prompt on average; between two scorings of the same prompt 6.5% of examples change outcome on average (up to 16.6%), and the pilot and present scorings of the same prompt differ by up to 7.0 points. This is larger than the 2% assumed in the first submission and is the reason a 25-question validation split is noisy. For 200 test examples the minimum detectable paired effect at 80% power and $\alpha = 0.05$ is 5.0 points if only serving noise separates two prompts (paired discordance 6.5%), and 6.3 to 8.9 points if two different prompts disagree on 10-20% of examples, as candidates here do. The strict gains of the strong refiner (+30, +29, +12, +11 points) are above this floor; effects of a few points are not detectable at these test sizes.

5.8 DOES THE RUBRIC JUDGE MEASURE ANYTHING? (RQ4)

The 1.7B judge gives a parseable total for 93% of prompts and scores them 7.2 points higher than the reference judge on average; its totals for the same unchanged P0 vary with a standard deviation of 4.7 points across sampled rounds. Both judges' totals correlate only weakly with accuracy (median within-cell Spearman 0.27 and 0.27 with test accuracy), and the two judges agree with each other at 0.38. The rubric score is therefore at best a weak signal of prompt quality. No human ratings were collected; the reference judge is a stronger model, not a person, so this analysis bounds the judge's usefulness but does not validate the rubric against human judgement.

5.9 EFFECT OF THE ORIGINAL DEFECT

The original loop accepted the first round's outcome whatever it scored. With the 1.7B refiner and nine cells (MPIR seed 1, independent variant, oracle-scored candidates), applying that rule gives a mean test-accuracy change of -13.6 points relative to P0 when a fallback or echo round returns P0, as the pilot's code did: the prompt changed in 4 of 9 cells and was worse than P0 in 4, better in 0. Under the literal reading of protocol G6 (the first validated candidate is accepted unconditionally) the mean change is -29.2 points. Two cells account for most of either loss, the repaired loop changes accuracy by -0.9 points on the same candidates, and the figure applies to this refiner only; its effect with the original GPT-4o refiner is unknown, because that configuration was not rerun.

5.10 DECISION RULES

6. DISCUSSION

6.1 WHAT THE EVIDENCE SUPPORTS

With Qwen3-1.7B as target and refiner, the rubric-guided layer, as specified, did not improve the prompts of any of three optimizers on any of nine tasks: the final prompt equalled P0 in 79 of 80 runs, and by the pre-specified rule the outcome is null (Section 5.3). The refinement rounds were made distinct by sampling and history, the failure is not caused by the inconsistent rubric text, and a rubric-free gated rewrite does at least as well (Section 5.5). The main reason is the refiner: most rounds never produce a usable candidate, and the unprotected candidates lose the answer-format line, so under strict extraction they score zero. With the format line appended the refiner still reaches validation in a minority of rounds and 5 of 36 validated candidates beat the baseline (Section 5.2). The null is therefore a result about a 1.7B model's ability to follow a long refinement instruction, and it says little about rubric-guided refinement by a capable model.

The evidence on a more capable refiner is mixed and exploratory. A much larger open-weight refiner produced significant strict gains in four cells in two MPIR seeds, and a strong closed refiner in one. Most of the strict gains are, however, repairs of the answer tag on base prompts that reason about as well as before; under lenient extraction they reverse, because the long structured rewrite makes the 1.7B target reason worse, and a rubric-free gated rewrite by the same refiner reaches gains of the same kind. The one gain that survives both metrics, hyperbaton with PromptWizard, is on a base prompt that opens with an irrelevant sentence, and it is a single cell. So the data do not show that the seven-criteria rubric helps; criterion-level ablations were not run, and

TABLE 10: Rubric totals (out of 35) assigned at temperature 0 to the base prompts and oracle-scored candidates (223 distinct prompts). Spearman correlations of the total with validation and with test accuracy are computed within each of the nine cells over the 275 oracle-scored candidates (median shown); the two judges' totals correlate at 0.38.

Judge	Totals parsed	Mean total	SD	Spearman, val.	Spearman, test
Qwen3-1.7B	93%	27.9	9.1	0.28	0.27
GPT-6 Luna (reference)	100%	20.7	8.2	0.27	0.27

TABLE 11: Outcome of the decision rules (Section 4.1).

Rule	Criterion	Observed	Outcome
D1	At least 15 of 45 cells win and mean at least +2.0 pts; null if at most 2 wins and |mean| at most 1.0	0 wins, mean -0.04	Null
D2, GPT-6 Luna	Mean +2.0 over 1.7B and at least 3 of 9 wins	+3.0; 1 win	Not met
D2, Qwen3-235B (post hoc)	Same	+8.7; 4 wins (lenient: -3.2; 1 win)	Met under strict only
D3	Best oracle candidate at least +3 pts in at least 3 of 9 cells although the gated final is not	0, 0, 1 cells (V1, V2, V3)	Not met

the original GPT-4o and GPT-3.5-turbo configuration was not repeated, so the first evaluation's per-task gains remain unexplained and should not be cited as evidence for MPIR.

A second lesson is about measurement. Answer extraction decides what the gate rewards: a gate that uses strict extraction accepts rewrites that fix the tag even when they harm reasoning, and exemplar-free baselines look far weaker than they are. Reporting a lenient metric next to the strict one exposed both effects.

6.2 PRACTICAL GUIDANCE, SCOPED TO WHAT WAS TESTED

The following points follow from these experiments, with Qwen3 models, nine BBH tasks and the optimizers listed; they are not general rules. (i) Before adding a refinement layer, inspect how answers are extracted and check a lenient metric: one worked example of the answer format moved strict accuracy of exemplar-free prompts by +19 and +42 points (APE, ProTeGi), and +2 and +10 points under lenient extraction. (ii) Do not accept a model's rewrite of a prompt without scoring the original prompt as a baseline; with a 1.7B refiner the uncorrected loop lowered accuracy by about 14 points on average in our nine cells (one MPIR seed). (iii) Gate on held-out data, but note that identical prompts scored twice differ on 6.5% of examples, so differences of a few points on a few hundred questions are noise. (iv) When the refining model is as small as the target, expect most rewrites to be unusable (echoes, missing delimiters, a dropped answer format) and monitor that rate; protect the answer-format line explicitly.

6.3 THREATS TO VALIDITY

6.3.1 Internal Validity

The serving stack is not deterministic: repeated scoring of one prompt spans 3.3 points on average (standard deviation 1.4), which the gate and the minimum detectable effect (5.0 points) take into account but cannot remove. The inference server was restarted several times and its memory settings were changed during the study; cached scores and generations were kept across restarts, so a few interrupted runs combine scorings taken under different settings, with differences of the order of the measured noise. The main grid has 80 runs instead of the planned 135, oracle scoring covers MPIR seed 1 only, the strong-refiner runs use one MPIR seed (a second for one refiner), and the format-protected variants cover only part of the nine Tier-1 seed-42 cells (Section 5.2). Two audit findings, the format asymmetry of V1 and V2 and a lenient parser that rejected bare letters, were found late and corrected after the main grid had run; the main grid was not re-run with protection. The follow-up runs and the protected variants were added after results were seen. The APE prompt for hyperbaton is a degenerate artifact of our implementation (Appendix A), and many APE and ProTeGi base prompts are weak, which limits what refining them can show.

6.3.2 External Validity

One target model family and size was used in every run. Tier 1 was chosen from a biased statistic and its tasks were also used while the rubric was developed; Tier 2 is a random draw of six tasks from 24, with one split seed. The strong-refiner experiment covers nine cells. The original closed-model configuration, including GPT-4o as refiner, was not rerun; GPT-6 Luna is a different model and not a replication of it.

6.3.3 Construct and Conclusion Validity

Strict answer extraction partly measures output formatting, and the lenient metric is an approximation. The rubric and its judge are weakly related to accuracy (Section 5.8), no human ratings were collected, and the rubric's criteria were not ablated. The statistics are descriptive by design: with identical prompts in most cells, tests of differences would be uninformative, the nine pilot cells share three tasks and three optimizers and are not independent, and the cell-level wins of the strong refiners carry no multiplicity correction.

7. CONCLUSION

We re-tested a rubric-guided refinement layer for automatically optimized prompts after repairing its refinement loop and fixing the decision rules in advance. With a 1.7B model refining its own prompts the layer produced no improvement in 45 cells (one of 80 runs accepted a candidate), mainly because that refiner rarely produces a usable, format-preserving candidate. A much larger refiner obtained strict-metric gains in several cells, but most reverse under lenient scoring and a rubric-free rewrite reaches gains of the same kind, so the rubric is not shown to be what helps; one gain, on a single task, holds under both metrics. With a 1.7B refiner the uncorrected loop would have reduced accuracy by about 14 points (its effect with the original GPT-4o refiner is unknown), strict answer extraction accounts for much of the apparent weakness of exemplar-free baselines, and identical prompts do not score identically on this serving stack. These conclusions are limited to Qwen3-based targets, nine tasks and the optimizers tested, and the first evaluation's GPT-4o results remain unverified. Code, per-round logs and generations are public.

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9. ETHICAL STATEMENT

This study did not involve human participants, human data, or animal subjects, and did not require ethical approval.

10. DECLARATION OF COMPETING INTEREST

The authors declare that they have no conflicts of interest.

11. DATA AVAILABILITY STATEMENT

The datasets used in this study are drawn from the publicly available Big-Bench Hard benchmark. Code, prompts, the revision protocol, per-round logs, generations, and analysis outputs are openly available at https://github.com/millennials92/meta_prompted_instruction_refinement.

12. AUTHOR CONTRIBUTION STATEMENT

Linh Nguyen: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing - original draft, Visualization. Quang-Vinh Dang: Conceptualization, Methodology, Supervision, Writing - review & editing, Project administration. Minh Ngoc Dinh: Methodology, Validation, Writing - review & editing. Thuy Nguyen: Investigation, Validation, Writing - review & editing.

APPENDIX A PROMPT EXAMPLES

A.1 BASE PROMPTS PRODUCED BY A 1.7B-DRIVEN OPTIMIZER

These are prompts exactly as saved by the pipeline (seed 42), not cleaned. They show the quality of the upstream, 1.7B-driven baselines that MPIR was asked to refine, which limits how informative refining them can be. PromptWizard's prompt for penguins_in_a_table opens with a detached question that is not a task instruction; the APE prompt for hyperbaton is APE's own instruction-induction meta-prompt, not a task instruction, because the search selected a candidate that echoed the induction template; the APE and ProTeGi prompts for the other two tasks are the seed instruction "Let's think step by step." plus the answer-format line.

```
How could I determine the accuracy of
my solution to this problem?

[Question] Here is a table where the
first line is a header and
each subsequent line is a penguin: name,
age, height (cm),
weight (kg) Louis, 7, 50, 11 Bernard,
5, 80, 13 Vincent, 9, 60, 11 ...
What is the name of the 60 cm tall penguin?
Options: (A) Louis (B) Bernard (C) Vincent
(D) Gwen (E) James
[Answer] ... Therefore, the name of the
60 cm tall penguin is Vincent.
**Final Answer:** (C) Vincent <ANS_START>(C)<ANS
END>
[... two further worked examples omitted
...]
Wrap the final answer between <ANS_START>
and <ANS_END>
```

Figure A.1. PromptWizard's saved prompt for penguins_in_a_table (abridged).

```
I gave a friend an instruction. Based
on the instruction they
produced the following input-output pairs:
[Question] Which sentence has the correct
adjective order: ...
[Answer] (B)
[... further input-output pairs omitted
...]
What was the instruction I gave?
Wrap the final answer between <ANS_START>
and <ANS_END>
```

Figure A.2. APE's saved prompt for hyperbaton (abridged): an unextracted induction meta-prompt, not a task instruction.

A.2 REFERENCE PROMPTS

The zero-shot condition uses the BBH authors' task description plus "Let's think step by step" and the answer-format line. The expert few-shot condition appends the authors' three worked chain-of-thought exemplars [20], each followed by a bare <ANS_START>(X)<ANS_END> tag. This tag is the property that exemplar-free prompts lack (Section 5.1).

```
Q: Which of the following is a humorous
edit of this artist or movie name: 'whitesnake'?
Options: (A) whitesnape (B) whitesnapke
(C) whitesnuake (D) mwhitesnake
A: Let's think step by step. ... So the
answer is (A).
<ANS_START>(A)<ANS_END>
```

Figure A.3. First expert exemplar for ruin_names, abridged. The same exemplar is the single format exemplar used in the intervention of Section 5.1.

A.3 FREE-REWRITE CONTROL PROMPT

```
You are given a task-solving prompt generated
for a Big-Bench Hard task.
Rewrite the prompt to improve clarity,
readability, and instruction organization
while
preserving the original task meaning.
Preserve the original task, answer format,
and examples.
Do not use any prompt-evaluation rubric.
Output only the rewritten task-solving
prompt.
Original prompt: <START> {prompt} <END>
```

Figure A.4. Meta-prompt of the free-rewrite control (abridged).

APPENDIX B ECHO FAILURE TRANSCRIPTS

Two ways in which a small refiner failed silently, found by inspecting saved outputs after an anomalous accuracy reading in the pilot. Both are well-formed <START>/<END> responses with no usable task instruction. They are now caught by the echo guard (Section 3.4) and, independently, by the validation gate.

```
<START>
I gave a friend an instruction. Based
on the instruction they produced the
following
input-output pairs: [Question] Which
sentence has the correct adjective order:
...
The task is to determine the correct
adjective order in a sentence ...
<END>
```

Figure B.1. hyperbaton / APE: the refiner echoed APE's induction template; no answer-format line, so validation accuracy fell to zero (abridged).

```
<START>
### Task Overview You are a Senior Prompt
Engineer ... ### Prompt Under Evaluation
...
### Seven-Criteria Evaluation Rubric
...
<END>
```

Figure B.2. ruin_names / PromptWizard: the refiner echoed the evaluation meta-prompt itself; the round scored 0.0 (abridged).

```
P0 (ProTeGi, hyperbaton): Let's think
step by step. Wrap the final answer between
<ANS_START> and <ANS_END>
Accepted candidate (Qwen3-235B refiner,
round 7; validation 0.52 vs 0.16, test
39.5% vs 10.5%):
You are a Senior Reasoning Assistant
... ### Instructions 1. Step Back ...
2. Reason Step-by-Step ...
5. Follow Format Exactly ... 7. Final
Answer: On a new line, wrap only the
final answer with
<ANS_START> and <ANS_END>. ### Output
Format Examples [Question] A bat and
a ball cost $1.10 ... [Answer] ...
```

Figure B.3. An accepted candidate from the strong refiner (abridged). It adds a role, step-back and numbered steps, and also an explicit instruction to wrap the final answer on its own line and a worked output example, which repairs the answer tag; the example is generic, not about adjective order. The strict accuracy of this prompt is higher than that of P0 (39.5% against 10.5%) but its lenient accuracy is lower (16% against 58%, Section 5.6).

APPENDIX C PIPELINE DEFECTS AND SERVING CONFIGURATION

Defects found while running the pipeline against a live model. Magnitudes are the symptoms recorded at the time. Items 1-8 were fixed before any number reported in Section 5; items 9-13 were found during this revision, items 12 and 13 late, after the main grid had run. Earlier pre-pilot fixes (a per-seed data-split collision, an unseeded random generator, mismatched scoring minibatches in APE, and swallowed API errors) are recorded in the repository's REBUILD.md.

APPENDIX D THE ORIGINAL EVALUATION (EXPLORATORY; VALIDATION LEAK)

The first evaluation used GPT-3.5-turbo as target and GPT-4o as refiner on all BBH tasks. It is reproduced here for completeness, because the Introduction refers to it. Its validation stage scored candidates on the optimizer's own training examples and its loop never scored the unrefined prompt (Section 3.2), single runs at temperature 0. The numbers are exploratory, are not evidence for MPIR, and were not used to compute any result in Section 5. The Gemini 3.5 Flash-Lite check used the same model as target and refiner, with the same defects; its baseline already averaged 92%, leaving little headroom, and it was not analysed further. The three pilot tasks ranked 1, 3 and 6 of 23 by the original PW+MPIR gain (hyperbaton +21.8,

TABLE 12: Defects found in the pipeline, their observed symptom and the fix.

#	Defect	Observed symptom	Fix
1	Response truncated by an under-provisioned context	PromptWizard prompts left too little context: responses cut mid-sentence, 0/50 on a task	Explicit max new tokens (1024) and context 8192
2	Answer normalization gap	Bare letter 'A' scored wrong against gold '(A)'	Strip parentheses and case on both sides
3	Delimiter convention absent from expert exemplars	Expert few-shot scored exactly 0.0 on hyperbaton (whole response wrapped)	Append a bare tag after each exemplar's conclusion
4	Missing delimiters crashed the loop	IndexError ended every remaining condition of a task	Fallback to the unrefined prompt (now logged as an outcome)
5	Echo of the rubric meta-prompt	ruin_names / PromptWizard candidate scored 0.0	Marker-phrase echo guard
6	Echo of APE's induction template; unscored baseline	hyperbaton / APE candidate scored 0.0 and was accepted because the incumbent score started at minus infinity	Score P0 first; accept only on strict improvement
7	Evaluation template mismatch	Same prompt text scored 0.07 through one template and 0.515 through another (trailing whitespace changes greedy decoding)	Score every condition through the base optimizer's own template
8	Search-time scoring omitted the answer format	APE/ProTeGi candidates scored near 0 during search, so selection ran on noise	Bake the answer-format line into scored instructions
9	Rubric text inconsistent	'five criteria', 'all 6 criteria', 'X/25' for a seven-criterion, 35-point rubric	Corrected text for V1-V3; original kept as V2L
10	Refinement rounds not independent	Same input, temperature 0, seven times	Independent sampling and chained variants (Section 3.2)
11	APE search selected a degenerate prompt	hyperbaton P0 is APE's own induction template	Left in place and disclosed; same code as the pilot
12	Candidates of V1 and V2 scored without the answer-format line	The refinement request does not ask to keep it; 93-95% of validated 1.7B candidates lack it and score exactly zero under strict extraction	Format-protected variants V1P and V2P (Section 3.2); asymmetry reported
13	Lenient parser rejected a bare letter inside the	Strict accepted what lenient rejected	Lenient now accepts everything strict accepts

TABLE 13: Serving configuration.

Item	Value
Inference engine	vLLM 0.28.0 (V1 engine, model runner V2 disabled, flashinfer sampler disabled), PyTorch 2.13.0+cu132
Target and default refiner	Qwen/Qwen3-1.7B, revision 70d244c, bf16, no quantization
Capability-experiment refiners	qwen/qwen3-235b-a22b-2507 and openai/gpt-6-luna (reasoning effort minimal) through the OpenRouter API, 2026-10-20 and 21, temperature 0.7, 3000 new tokens; no local server
Context and limits	max model length 8192; max new tokens 1024 (target), 1500 / 3000 (meta calls); thinking disabled
Decoding	target: temperature 0; refiner: temperature 0.7 (0 for V0); seeds recorded per call
Server flags	–gpu-memory-utilization 0.80-0.86 –max-num-seqs 64-128 –seed 0; the server was restarted several times and its memory settings changed (fp8 KV cache was tried and not usable without nvcc; prefix caching was off for one period); cached scores and generations were kept across restarts
Hardware	NVIDIA RTX 5000 Ada Generation Laptop GPU, 16 GB, driver 596.58; WSL2 (Ubuntu 24.04, kernel 5.15.167.4) on Windows 11
Non-determinism	Continuous batching; identical prompts are re-scored to measure it (Section 5.7)

penguins_in_a_table +7.6, ruin_names +5.8 points). Selecting tasks on the maximum of a noisy, leaked statistic invites winner's-curse effects, which is why Tier 2 of the revised design draws tasks at random.

APPENDIX E ROUND-LEVEL RECORD OF THE PRIMARY VARIANT

Per cell (task / optimizer, seed 42) and round, the outcome of the chained variant V2 with MPIR seed 1 (F fallback, E echo, D duplicate, V validated with its validation accuracy), then the baseline validation accuracy of P0 and the judge's rubric total out of 35 at round 1. Table 16 gives the same for the independent variant V1. Full logs for every variant, seed and round, including candidate texts and judge outputs, are in results/revision/rounds.

APPENDIX F PROTOCOL AND DEVIATIONS

The decision rules of Section 4.1 were fixed in REVISION_PROTOCOL.md before any revised experiment ran (commit 8fd1397, 2026-10-19). The amendments below are recorded in that file; none is an entry in a registry. Amendments 1 to 7 were made before the results of the affected experiment were

TABLE 14: Original GPT-3.5-turbo evaluation (accuracy %, single run, leaked validation split) and Gemini 3.5 Flash-Lite check. PW = PromptWizard; Diff = PW+MPIR minus PW (points). Exploratory only.

Task	Zero-shot	Expert	PW	PW+MPIR	Diff	APE	APE+MPIR	ProTeGi	ProTeGi+MPIR	Gemini PW	Gemini MPIR
hyperbaton	74.6	84.4	62.2	84.0	+21.8	82.7	85.8	80.9	88.9	100	100
disambiguation_qa	55.1	69.3	61.7	62.2	+0.5	69.8	68.0	73.8	67.6	77	80
causal_judgement	53.7	59.9	59.3	56.8	-2.5	58.0	58.6	61.1	58.6	70	75
date_understanding	55.6	80.0	71.1	74.2	+3.1	87.6	84.4	88.0	85.8	95	96
penguins_in_a_table	45.5	74.4	75.0	82.6	+7.6	81.0	84.3	79.3	76.9	100	98
boolean_expression	81.8	91.6	92.0	94.2	+2.2	90.7	93.8	90.7	92.9	100	100
object_counting	59.1	88.0	91.5	92.4	+0.9	92.0	93.3	88.4	93.3	100	100
word_sorting	82.6	65.3	80.0	81.3	+1.3	69.3	80.0	68.4	66.7	92	92
logical_deduction	45.3	59.4	44.9	50.8	+5.9	61.9	59.7	60.7	59.3	99	96
salient_translation_error_detection	44.0	54.7	50.7	52.0	+1.3	55.1	54.7	52.9	49.3	75	74
geometric_shapes	29.3	64.4	57.8	49.3	-8.5	62.7	61.3	61.8	60.9	86	83
snarks	45.0	72.5	63.3	65.4	+2.1	61.4	68.6	75.8	73.2	90	90
temporal_sequences	33.3	17.3	44.4	38.2	-6.2	26.7	86.7	84.0	86.2	100	100
web_of_lies	47.5	80.9	52.4	52.9	+0.5	80.0	73.3	81.3	80.9	100	100
navigate	49.3	93.8	65.3	68.0	+2.7	95.6	92.9	95.6	97.8	100	99
reasoning_about_colored_objects	54.2	83.1	56.0	68.4	+12.4	82.7	82.7	82.7	83.6	100	100
sports_understanding	78.2	93.8	79.6	86.2	+6.6	95.1	96.4	93.3	94.7	91	86
multistep_arithmetic_two	33.3	84.9	51.1	51.5	+0.4	86.2	84.9	83.6	82.7	97	94
ruin_names	54.7	69.3	66.2	72.0	+5.8	58.7	68.4	58.7	77.8	88	87
movie_recommendation	66.2	79.1	73.3	66.7	-6.6	76.9	79.1	76.0	80.9	95	96
formal_fallacies	53.8	51.6	53.3	53.3	+0.0	52.4	52.9	55.1	56.9	99	99
dyck_languages	1.7	16.9	14.2	12.9	-1.3	24.4	34.2	20.9	32.4	72	75
tracking_shuffled_objects	22.4	63.8	69.8	65.5	-4.3	63.0	58.2	61.6	60.9	100	100
Average	50.7	69.5	62.4	64.4	+2.0	70.2	74.0	72.8	74.3	92	92

TABLE 15: Round-level outcomes, variant V2, MPIR seed 1 (an asterisk marks an accepted candidate).

Cell	P0 val	R1	R2	R3	R4	R5	R6	R7	Judge /35
hyperbaton / promptwizard	0.72	E	E	E	E	E	V 0.72	F	31
hyperbaton / ape	0.64	F	E	E	F	F	V 0.00	F	33
hyperbaton / protegi	0.16	V 0.00	F	V 0.00	F	V 0.00	F	F	16
ruin_names / promptwizard	0.76	E	E	E	F	E	E	E	28
ruin_names / ape	0.28	V 0.00	F	V 0.00	V 0.00	F	F	F	17
ruin_names / protegi	0.08	V 0.00	V 0.00	F	F	F	V 0.00	V 0.00	17
penguins_in_a_table / promptwizard	0.84	V 0.00	F	F	F	F	F	F	31
penguins_in_a_table / ape	0.24	V 0.00	F	V 0.00	F	F	F	F	16
penguins_in_a_table / protegi	0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	17

TABLE 16: Round-level outcomes, variant V1, MPIR seed 1 (an asterisk marks an accepted candidate).

Cell	P0 val	R1	R2	R3	R4	R5	R6	R7	Judge /35
hyperbaton / promptwizard	0.72	E	E	E	V 0.68	V 0.72	V 0.76*	E	29
hyperbaton / ape	0.64	V 0.00	E	E	E	E	E	E	32
hyperbaton / protegi	0.16	F	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	F	17
ruin_names / promptwizard	0.76	E	E	E	V 0.00	E	E	E	31
ruin_names / ape	0.28	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	16
ruin_names / protegi	0.08	V 0.00	F	V 0.00	V 0.00	F	V 0.00	V 0.00	17
penguins_in_a_table / promptwizard	0.84	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	30
penguins_in_a_table / ape	0.24	F	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	V 0.00	15
penguins_in_a_table / protegi	0.00	V 0.00	V 0.00	V 0.00	V 0.00	F	V 0.00	V 0.00	16

inspected; amendments 8 and 9 were made after results were seen and are exploratory; V1P and V2P were stopped before all nine cells finished.

- 1. 2026-10-20, compute. Serving stack vLLM 0.28.0 under WSL2 with measured throughput of about 1.2k tokens/s; oracle test scoring of every validated candidate for MPIR seed 1 only; one validation score per candidate in the main grid G1; V2L with one seed and no oracle scoring.
- 2. 2026-10-20, sampling. Refiner sampling uses temperature 0.7 without top-p (the protocol text said top-p 0.95), because top-p was about three times slower on this stack.
- 3. 2026-10-20, grouping. Tier-1 seed-42 cells are run only in the mechanism group G2, which contains the V2 runs with oracle scoring; G1 covers the other 36 cells.
- 4. 2026-10-20, refiners for D2. Local 4B and 8B refiners were not run; the strongest-refiner role of D2 went to GPT-6 Luna and Qwen3-235B-A22B through OpenRouter (Section 4.1).
- 5. 2026-10-20 and 21, scale. The main grid has 80 runs instead of the planned 135; MPIR seeds beyond 1 were cut after seed 1 was complete.
- 6. 2026-10-20, pooling. Cell-level differences pool the MPIR seeds of a cell by averaging per-example outcomes (Section 4.1).
- 7. 2026-10-20, optimizer implementations. No change to APE, ProTeGi or PromptWizard code or hyperparameters between the pilot and the extra

seeds and tasks.

- 8. 2026-10-21, exploratory. After the Qwen3-235B refiner showed strict-metric gains, the rubric-free gated rewrite with that refiner and a second MPIR seed were added (Section 5.6.1).
- 9. 2026-10-21, exploratory. After an audit found that V1 and V2 candidates are scored without the answer-format line, the format-protected variants V1P and V2P were added, and the lenient parser was corrected to accept everything the strict parser accepts (Section 5.2).

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REFERENCES

[1] D. Vilar, M. Freitag, C. Cherry, J. Luo, V. Ratnakar, and G. Foster, "Prompting PaLM for translation: Assessing strategies and performance," in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2023, pp. 15 406–15 427.

[2] T. Goyal, J. J. Li, and G. Durrett, "News summarization and evaluation in the era of GPT-3," arXiv preprint arXiv:2209.12356, 2022.

[3] E. Kamalloo, N. Dziri, C. Clarke, and D. Rafiei, "Evaluating open-domain question answering in the era of large language models," in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2023, pp. 5591–5606.

[4] L. Qin, Q. Chen, X. Feng, Y. Wu, Y. Zhang, Y. Li, M. Li, W. Che, and P. S. Yu, "Large language models meet NLP: A survey," Frontiers of Computer Science, 2025, arXiv:2405.12819.

[5] P. Colombo, T. P. Pires, M. Boudiaf, D. Culver, R. Melo, C. Corro, A. F. T. Martins, F. Esposito, V. L. Raposo, S. Morgado, and M. Desa, "Saullm-7b: A pioneering large language model for law," arXiv preprint arXiv:2403.03883, 2024.

[6] X. Zhang, C. Tian, X. Yang, L. Chen, Z. Li, and L. R. Petzold, "AlpaCare: Instruction-tuned large language models for medical application," arXiv preprint arXiv:2310.14558, 2023.

[7] F. Errica, D. Sanvito, G. Siracusano, and R. Bifulco, "What did I do wrong? Quantifying LLMs' sensitivity and consistency to prompt engineering," in Proc. 2025 Conf. Nations of the Americas Chapter of the Assoc. Comput. Linguistics: Human Language Technologies (Volume 1: Long Papers), 2025, pp. 1543–1558.

[8] M. Sclar, Y. Choi, Y. Tsvetkov, and A. Suhr, "Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting," in The Twelfth International Conference on Learning Representations, 2024.

[9] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in Proceedings of the 36th International Conference on Neural Information Processing Systems (Article 1800), 2022.

[10] A. Kong, S. Zhao, H. Chen, Q. Li, Y. Qin, R. Sun, X. Zhou, E. Wang, and X. Dong, "Better zero-shot reasoning with role-play prompting," in Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (Volume 1: Long Papers), 2024, pp. 4099–4113.

[11] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, D. Amodei et al., "Language models are few-shot learners," in Proceedings of the 34th International Conference on Neural Information Processing Systems (Article 159), 2020.

[12] J. D. Zamfirescu-Pereira, R. Y. Wong, B. Hartmann, and Q. Yang, "Why Johnny can't prompt: How non-AI experts try (and fail) to design LLM prompts," in Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems, 2023.

[13] L. Reynolds and K. McDonell, "Prompt programming for large language models: Beyond the few-shot paradigm," in Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems, 2021.

[14] Y. Zhou, A. I. Muresanu, Z. Han, K. Paster, S. Pitis, H. Chan, and J. Ba, "Large language models are human-level prompt engineers," in The Eleventh International Conference on Learning Representations, 2023.

[15] R. Pryzant, D. Iter, J. Li, Y. T. Lee, C. Zhu, and M. Zeng, "Automatic prompt optimization with "gradient descent" and beam search," in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, 2023.

[16] Q. Guo, R. Wang, J. Guo, B. Li, K. Song, X. Tan, G. Liu, J. Bian, and Y. Yang, "Connecting large language models with evolutionary algorithms yields powerful prompt optimizers," in The Twelfth International Conference on Learning Representations, 2024.

[17] C. Yang, X. Wang, Y. Lu, H. Liu, Q. V. Le, D. Zhou, and X. Chen, "Large language models as optimizers," in The Twelfth International Conference on Learning Representations, 2024.

[18] E. Agarwal, R. Magazine, J. Singh, V. Dani, T. Ganu, and A. Nambi, "Promptwizard: Optimizing prompts via task-aware, feedback-driven self-evolution," in Findings of the Association for Computational Linguistics: ACL 2025, 2025, pp. 19 974–20 003.

[19] K. Ramnath, K. Zhou, S. Guan, S. S. Mishra, X. Qi, Z. Shen, S. Wang, S. Woo, S. Jeoung, Y. Wang, H. Wang, H. Ding, Y. Lu, Z. Xu, Y. Zhou, B. Srinivasan, Q. Yan, Y. Chen, H. Ding, P. Xu, and L. L. Cheong, "A systematic survey of automatic prompt optimization techniques," pp. 33 066–33 098, 2025.

[20] M. Suzgun, N. Scales, N. Schärli, S. Gehrmann, Y. Tay, H. W. Chung, A. Chowdhery, Q. Le, E. Chi, D. Zhou, and J. Wei, "Challenging big-bench tasks and whether chain-of-thought can solve them," in Findings of the Association for Computational Linguistics: ACL 2023, 2023, pp. 13 003–13 051.

[21] P. Liu, W. Yuan, J. Fu, Z. Jiang, H. Hayashi, and G. Neubig, "Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing," ACM Computing Surveys, vol. 55, 2023.

[22] Q. Dong et al., "A survey on in-context learning," in Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, 2024, pp. 1107–1128.

[23] S. Min, X. Lyu, A. Holtzman, M. Artetxe, M. Lewis, H. Hajishirzi, and L. Zettlemoyer, "Rethinking the role of demonstrations: What makes in-context learning work?" in Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, 2022, pp. 11 048–11 064.

[24] T. Kojima, S. S. Gu, M. Reid, Y. Matsuo, and Y. Iwasawa, "Large language models are zero-shot reasoners," in Proceedings of the 36th International Conference on Neural Information Processing Systems (Article 1613), 2022.

[25] X. Wang and D. Zhou, "Chain-of-thought reasoning without

prompting," in Proceedings of the 38th International Conference on Neural Information Processing Systems, 2024.
[26] B. Wang, S. Min, X. Deng, J. Shen, Y. Wu, L. Zettlemoyer, and H. Sun, "Towards understanding chain-of-thought prompting: An empirical study of what matters," in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2023, pp. 2717–2739.
[27] J. Kim, N. Yang, and K. Jung, "Persona is a double-edged sword: Mitigating the negative impact of role-playing prompts in zero-shot reasoning tasks," arXiv preprint arXiv:2408.08631, 2024, arXiv preprint.
[28] M. Zheng, J. Pei, L. Logeswaran, M. Lee, and D. Jurgens, "When "a helpful assistant" is not really helpful: Personas in system prompts do not improve performances of large language models," in Findings of the Association for Computational Linguistics: EMNLP 2024, 2024, pp. 15 126–15 154.
[29] H. S. Zheng, S. Mishra, X. Chen, H.-T. Cheng, E. H. Chi, Q. V. Le, and D. Zhou, "Take a step back: Evoking reasoning via abstraction in large language models," in The Twelfth International Conference on Learning Representations, 2024.
[30] J. Coda-Forno, M. Binz, J. X. Wang, and E. Schulz, "Cogbench: A large language model walks into a psychology lab," in Proceedings of the 41st International Conference on Machine Learning, 2024.
[31] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," Transactions of the Association for Computational Linguistics, vol. 12, pp. 157–173, 2024.
[32] Y. Yu, H. Jiang, X. Luo, Q. Wu, C.-Y. Lin, D. Li, Y. Yang, Y. Huang, and L. Qiu, "Mitigate position bias in llms via scaling a single hidden states channel," in Findings of the Association for Computational Linguistics: ACL 2025, 2025, pp. 6092–6111.
[33] Microsoft, "Prompt engineering techniques," Microsoft Foundry documentation, Microsoft Learn, 2026, page dated May 13, 2026. Accessed Oct. 20, 2026. Archived (earlier URL): https://web.archive.org/web/20251112071324/https://learn.microsoft.com/en-us/AZURE/AI-FOUNDRY/OPENAI/CONCEPTS/PROMPT-ENGINEERING. [Online]. Available: https://learn.microsoft.com/en-us/azure/foundry/openai/concepts/prompt-engineering
[34] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, "Self-consistency improves chain of thought reasoning in language models," in The Eleventh International Conference on Learning Representations, 2023.
[35] O. Khattab, A. Singhvi, P. Maheshwari, Z. Zhang, K. Santhanam, S. Vardhamanan, S. Haq, A. Sharma, T. T. Joshi, H. Moazam, H. Miller, M. Zaharia, and C. Potts, "Dspy: Compiling declarative language model calls into state-of-the-art pipelines," in The Twelfth International Conference on Learning Representations, 2024.
[36] M. Yuksekgonul, F. Bianchi, J. Boen, S. Liu, P. Lu, Z. Huang, C. Guestrin, and J. Zou, "Optimizing generative ai by backpropagating language model feedback," Nature, vol. 639, pp. 609–616, 2025.
[37] A. Madaan, N. Tandon, P. Gupta, S. Hallinan, L. Gao, S. Wiegreffe, U. Alon, N. Dziri, S. Prabhumoye, Y. Yang, S. Gupta, B. P. Majumder, K. Hermann, S. Welleck, A. Yazdanbakhsh, and P. Clark, "Self-refine: Iterative refinement with self-feedback," in Thirty-seventh Conference on Neural Information Processing Systems, 2023.
[38] N. Shinn, F. Cassano, A. Gopinath, K. R. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in Thirty-seventh Conference on Neural Information Processing Systems, 2023.
[39] W. Cui, J. Zhang, Z. Li, H. Sun, D. Lopez, K. Das, B. A. Malin, and S. Kumar, "Heuristic-based search algorithm in automatic instruction-focused prompt optimization: A survey," in Findings of the Association for Computational Linguistics: ACL 2025, 2025, pp. 22 093–22 111.
[40] Q. Ye, M. Ahmed, R. Pryzant, and F. Khani, "Prompt engineering a prompt engineer," in Findings of the Association for Computational Linguistics: ACL 2024, 2024, pp. 355–385.
[41] M. Yang, M. Li, Y. Li, Z. Chen, C. Gao, J. Zhang, Y. Li, and F. Feng, "Dual-phase accelerated prompt optimization," in Findings of the Association for Computational Linguistics: EMNLP 2024, 2024, pp. 12 163–12 173.
[42] K. Mayilvaghanan, V. Nathan, and A. Kumar, "PROPEL: Prompt optimization with expert priors for small and medium-sized LLMs," in Proceedings of the 4th International Workshop on Knowledge-Augmented Methods for Natural Language Processing, 2025, pp. 272–302.
[43] A. Chakma, M. Surdeanu, and E. Blanco, "Two-stage prompt optimization for few-shot relation extraction: From reasoning-guided search to gradient-guided refinement," arXiv preprint arXiv:2606.29639, 2026.
[44] L. A. Agrawal, S. Tan, D. Soylu, N. Ziems, R. Khare, K. Opsahl-Ong, A. Singhvi, H. Shandilya, M. J. Ryan, M. Jiang, C. Potts, K. Sen, A. G. Dimakis, I. Stoica, D. Klein, M. Zaharia, and O. Khattab, "GEPA: Reflective prompt evolution can outperform reinforcement learning," International Conference on Learning Representations (ICLR), Oral, 2026.
[45] M. Singh, V. Yadav, and E. Blanco, "Error taxonomy-guided prompt optimization," arXiv preprint arXiv:2602.00997, 2026.
[46] Y. Liu, J. Xu, L. L. Zhang, Q. Chen, X. Feng, Y. Chen, Z. Guo, Y. Yang, and P. Cheng, "Beyond prompt content: Enhancing llm performance via content-format integrated prompt optimization," arXiv preprint arXiv:2502.04295, 2025.
[47] A. Nair, A. Banerjee, L. Mombaerts, M. Hagen, and T. Borogovac, "Tournament of prompts: Evolving LLM instructions through structured debates and Elo ratings," arXiv preprint arXiv:2506.00178, 2025.
[48] H.-C. Lin, B. M. Ruppik, C. van Niekerk, C.-H. Shen, M. Heck, N. Lubis, R. Vukovic, S. Feng, and M. Gasic, "Prompt reinforcing for long-term planning of large language models," arXiv preprint arXiv:2510.05921, 2025.
[49] S. Das, R. Kamoi, B. Pang, Y. Zhang, C. Xiong, and R. Zhang, "GReaTer: Gradients over reasoning makes smaller language models strong prompt optimizers," in The Thirteenth International Conference on Learning Representations, 2025.
[50] J. Shim, D. W. Kim, S. W. Kim et al., "POaaS: Minimal-edit prompt optimization as a service to lift accuracy and cut hallucinations on on-device sLLMs," arXiv preprint arXiv:2603.16045, 2026.
[51] Á. Pándy, R. Lakatos, and A. Hajdu, "Error-driven prompt optimization for arithmetic reasoning," IEEE Access, vol. 14, pp. 62 570–62 583, 2026.
[52] S. Badhe and D. Shah, "Prompt-level distillation: A non-parametric alternative to model fine-tuning for efficient reasoning," in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026), 2026, pp. 2131–2147.
[53] D. Shah, S. Badhe, N. Kathrotia, and P. Tiwari, "CROP: Token-efficient reasoning in large language models via regularized prompt optimization," arXiv preprint arXiv:2604.14214, 2026.
[54] Z. Zhu, H. Zhou, Z. Feng, T. Li, C. J. J. Deryl, M. L. Onn, G. W. Ng, and K. Mao, "Rethinking prompt optimizers: From prompt merits to optimization," arXiv preprint arXiv:2505.09930, 2025.
[55] S. M. Bsharat, A. Myrzakhan, and Z. Shen, "Principled instructions are all you need for questioning LLaMA-1/2, GPT-3.5/4," arXiv preprint arXiv:2312.16171, 2023.
[56] J. Huang, X. Chen, S. Mishra, H. S. Zheng, A. W. Yu, X. Song, and D. Zhou, "Large language models cannot self-correct reasoning yet," in The Twelfth International Conference on Learning Representations, 2024.
[57] Y. Zhang, M. Khalifa, L. Logeswaran, J. Kim, M. Lee, H. Lee, and L. Wang, "Small language models need strong verifiers to self-correct reasoning," in Findings of the Association

for Computational Linguistics: ACL 2024, 2024, pp. 15 637–15 653.
[58] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in Advances in Neural Information Processing Systems 36 (Datasets and Benchmarks Track), 2023.
[59] P. Wang, L. Li, L. Chen, Z. Cai, D. Zhu, B. Lin, Y. Cao, L. Kong, Q. Liu, T. Liu, and Z. Sui, "Large language models are not fair evaluators," in Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2024, pp. 9440–9450.
[60] A. Panickssery, S. R. Bowman, and S. Feng, "LLM evaluators recognize and favor their own generations," arXiv preprint arXiv:2404.13076, 2024.
[61] X. Wang, B. Ma, C. Hu, L. Weber-Genzel, P. Röttger, F. Kreuter, D. Hovy, and B. Plank, ""my answer is C": First-token probabilities do not match text answers in instruction-tuned language models," in Findings of the Association for Computational Linguistics: ACL 2024, 2024, pp. 7407–7416.
[62] C. Zheng, H. Zhou, F. Meng, J. Zhou, and M. Huang, "Large language models are not robust multiple choice selectors," in The Twelfth International Conference on Learning Representations, 2024.
[63] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché Buc, E. Fox, and H. Larochelle, "Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program)," Journal of Machine Learning Research, vol. 22, no. 164, pp. 1–20, 2021.
[64] J. Dodge, S. Gururangan, D. Card, R. Schwartz, and N. A. Smith, "Show your work: Improved reporting of experimental results," in Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, 2019, pp. 2185–2194.
[65] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Mohammadi Sepahvand, E. Raff, K. Madan, V. Voleti, S. Ebrahimi Kahou, V. Michalski, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in Proceedings of Machine Learning and Systems 3 (MLSys 2021), 2021.
[66] R. Dror, G. Baumer, S. Shlomov, and R. Reichart, "The hitchhiker's guide to testing statistical significance in natural language processing," in Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), 2018, pp. 1383–1392.
[67] D. Card, P. Henderson, U. Khandelwal, R. Jia, K. Mahowald, and D. Jurafsky, "With little power comes great responsibility," in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, 2020, pp. 9263–9274.
[68] S. Schulhoff, M. Ilie, N. Balepur, K. Kahadze, A. Liu, C. Si, Y. Li, A. Gupta, H. Han, S. Schulhoff, P. S. Dulepet, S. Vidyadhara, D. Ki, S. Agrawal, C. Pham, G. C. Kroiz, F. Li, H. Tao, P. Resnik et al., "The prompt report: A systematic survey of prompting techniques," arXiv preprint arXiv:2406.06608, 2024.
[69] P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, and A. Chadha, "A systematic survey of prompt engineering in large language models: Techniques and applications," arXiv preprint arXiv:2402.07927, 2024.
[70] L. Boonstra, "Prompt engineering," Google whitepaper, Kaggle, 2025, archived: https://web.archive.org/web/20260905081837/https://www.kaggle.com/whitepaper-prompt-engineering. Accessed Oct. 20, 2026. [Online]. Available: https://www.kaggle.com/whitepaper-prompt-engineering
[71] Anthropic, "Prompt engineering overview," Claude Developer Platform documentation, 2025, accessed Oct. 20, 2026. Page revised since the version used for the rubric; archived: https://web.archive.org/web/20261019210255/https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/overview. [Online]. Available: https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/overview
[72] OpenAI, "Best practices for prompt engineering with the OpenAI API," OpenAI Help Center, 2025, archived: https://web.archive.org/web/20260915104022/https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api. The live page could not be retrieved on Oct. 20, 2026. [Online]. Available: https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-the-openai-api
[73] A. Yang et al., "Qwen3 technical report," arXiv preprint arXiv:2505.09388, 2025.
[74] OpenAI, "GPT-6 Luna," Model listing on OpenRouter, 2026, accessed Oct. 20, 2026. [Online]. Available: https://openrouter.ai/openai/gpt-6-luna
[75] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. Gonzalez, H. Zhang, and I. Stoica, "Efficient memory management for large language model serving with PagedAttention," in Proceedings of the 29th Symposium on Operating Systems Principles, 2023, pp. 611–626.
[76] B. Efron, "Bootstrap methods: Another look at the jackknife," The Annals of Statistics, vol. 7, no. 1, pp. 1–26, 1979.
[77] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," Journal of the American Statistical Association, vol. 22, no. 158, pp. 209–212, 1927.

LINH NGUYEN Linh Nguyen is a Master's student in the Master of Artificial Intelligence program at RMIT University Vietnam, Ho Chi Minh City, Vietnam.

QUANG-VINH DANG Quang-Vinh Dang is a Senior Lecturer at British University Vietnam, Hung Yen, Vietnam.

MINH NGOC DINH Minh Ngoc Dinh is Dean of the School of Information Technology and Data Science, Millenia Education, Ho Chi Minh City, Vietnam.

THUY NGUYEN Thuy Nguyen is a Senior Lecturer at RMIT University Vietnam, Ho Chi Minh City, Vietnam.